Joy Ajwad

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EDUCATION

Wentworth Institute of Technology, Boston MA Expected August 2027

Bachelor of Science in Mechanical Engineering, Minor in Aerospace.
GPA: 3.67 | Dean's List - Fall 2023, Fall 2024, Spring 2025, Fall 2025.
Relevant Courses: Numerical Simulation & CFD, Design of Machine Elements, Engineering Dynamics, Fluid Mechanics, Thermodynamics I & II, Heat Transfer, Material Science, Differential Equations, Oral Communication.
Scholarships: Wentworth Presidential Merit Award, City of Cambridge Scholarship 2024, HRI Scholarship 2024, 2025, & 2026.

PROFESSIONAL EXPERIENCE

Vanderweil Engineers, Fire Protection Engineer Co-op
Boston, MA *Jan 2026 - Apr 2026*

- Independently developed fire protection and fire alarm system layouts, producing Revit-based BIM models for building systems, engineering drawings, and detailed support sheets.
- Performed code compliance analysis per NFPA 13, NFPA 72, IBC, and FM Global design standards.
- Supported multiple high-profile projects concurrently, including Class A high rises, industry-leading financial and institutional facilities, luxury hospitality, and mission-critical data centers.
- Conducted site visits documenting existing conditions of FA and FP systems against design drawings.
- Utilized Bluebeam to review drawings, track markups, and coordinate design updates with senior engineers.
- Coordinated with multidisciplinary teams to resolve clashes between fire protection and other building systems.
- Collaborated in engineering coordination meetings to review design progress, code requirements, and deliverables.

Integra Mission Critical (Critical Project Services), Project Engineer Co-op
Austin, TX *May 2025 - Aug 2025*

- Managed Fit-up and Core & Shell RFIs and submittals on a 235,000 sq. ft data center, for a Fortune 100 tech client.
- Led internal design meetings with EOR and AOR to resolve technical issues and prioritize critical design changes.
- Conducted site visits to monitor and verify construction progress and ensure alignment with project drawings.
- Performed install checklists on fuel tanks, electrical distribution systems, generators, & chillers, flagging priority issues.
- Implemented QA/QC checks on critical systems, identifying and addressing issues preemptively.
- Supported commissioning team prior to system energization and turnover in compliance with client standards.

TECHNICAL SKILLS

Software: SolidWorks, Autodesk Revit, Inventor, AutoCAD, Arduino, Multisim, Microsoft Office Suite, Bluebeam Revu.
Engineering: CAD/3D Modeling, Finite Element Analysis, R&D, Technical Drawing, BIM, Stress-Strain Analysis.
Certifications: Certified SolidWorks Associate (CSWA), Procore Engineering and Project Management Certification.

PROJECTS

Front - End Rebuild of Totaled Mazda 3 *Jan 2026*

- Performed extensive mechanical and structural evaluation of front-end assemblies, identifying engine, chassis, and body damage, and diagnosing engine non-start, overheating, and lean running conditions.
- Executed full front-end reconstruction using donor vehicle: Engineered bypass solution for lean operating conditions by precision tapping and installing custom vacuum lines and solenoids, restoring a proper engine air-fuel ratio. Identified source of overheating and performed an overhaul of cooling system and components.
- Successfully restored vehicle to normal operating conditions and validated system performance post-repair.

Single Stage Spur Gearbox *Nov 2025*

- Designed a single-stage spur gearbox for 4 hp at 2000 rpm, determining a 4.52:1 gear ratio within 0.5% of target.
- Performed detailed hand calculations for gear forces, loading factors, AGMA stress, and shaft fatigue; validated results using Autodesk Inventor simulations, verifying stress distribution and load conditions.
- Selected gear materials, lubrication, geometry, and mounting to ensure reliability under expected loads.
- Successfully developed a reliable, manufacturable gearbox meeting 4000-hour bearing life and 97.5% reliability.

LEADERSHIP/ACTIVITIES

- Active member of WIT SAE Baja Racing club, contributing to the engineering design, fabrication, and assembly of a competitive off-road Baja racing vehicle.
- Coordinated with CRLS community to create a dedicated Jummah prayer space, growing participation to 30+ members.